

## Week 1

### 1. Tic tac toe

Code:-

```
board = [' ' for _ in range(9)]
```

```
def print_board(board):  
    print(f"\n {board[0]} | {board[1]} | {board[2]} ")  
    print("-----")  
    print(f" {board[3]} | {board[4]} | {board[5]} ")  
    print("-----")  
    print(f" {board[6]} | {board[7]} | {board[8]} \n")
```

```
def check_winner(b, player):  
    win_conditions = [  
        [0, 1, 2], [3, 4, 5], [6, 7, 8],  
        [0, 3, 6], [1, 4, 7], [2, 5, 8],  
        [0, 4, 8], [2, 4, 6]  
    ]  
  
    return any(  
        all(b[i] == player for i in condition)  
        for condition in win_conditions  
    )
```

```
def is_full(b):  
    return ' ' not in b
```

```
def minimax(b, is_maximizing):  
    if check_winner(b, 'O'):  
        return 1
```

```
if check_winner(b, 'X'):
    return -1
```

```
if is_full(b):
    return 0
```

```
if is_maximizing:
    best_score = -float('inf')
```

```
for i in range(9):
    if b[i] == ' ':
        b[i] = 'O'
        score = minimax(b, False)
        b[i] = ' '

    best_score = max(score, best_score)
```

```
return best_score
```

```
else:
    best_score = float('inf')
```

```
for i in range(9):
    if b[i] == ' ':
        b[i] = 'X'
        score = minimax(b, True)
        b[i] = ' '

    best_score = min(score, best_score)
```

```
return best_score
```

```
def ai_move(b):
```

```
best_score = -float('inf')
move = None
```

```
for i in range(9):
    if b[i] == ' ':
        b[i] = 'O'
        score = minimax(b, False)
        b[i] = ' '
```

```
    if score > best_score:
        best_score = score
        move = i
```

```
b[move] = 'O'
```

```
print("You are X, AI is O.")
print("Choose positions from 0 to 8.")
```

```
print_board(board)
```

```
while True:
```

```
    try:
        pos = int(input("Enter move position (0-8): "))
```

```
    if pos < 0 or pos > 8:
        print("Please enter a number between 0 and 8.")
        continue
```

```
    if board[pos] == ' ':
        board[pos] = 'X'
```

```
    else:
        print("Invalid move! Try again.")
        continue
```

```
except ValueError:
    print("Please enter a valid number.")
    continue
if check_winner(board, 'X'):
    print_board(board)
    print("You win!")
    break

if is_full(board):
    print_board(board)
    print("It's a tie!")
    break

ai_move(board)

print("AI played:")
print_board(board)

if check_winner(board, 'O'):
    print("AI wins!")
    break

if is_full(board):
    print("It's a tie!")
    break
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Student/Desktop/NITYA/week 1 tic tack toe.py =====
You are X, AI is O.
Choose positions from 0 to 8.

  | |
  | |
  | |
  | |

Enter move position (0-8):
===== RESTART: C:/Users/Student/Desktop/NITYA/week 1 tic tack toe.py =====
You are X, AI is O.
Choose positions from 0 to 8.

  | |
  | |
  | |
  | |

Enter move position (0-8): 1
AI played:

  O | X |
  | |
  | |
  | |

Enter move position (0-8): 7
AI played:

  O | X |
  | O |
  | |
  | X |

Enter move position (0-8): 6
AI played:

  O | X |
  | O |
  | |
  X | X | O

AI wins!
>>>
```

## 2. Vacuum cleaner

Code:-

```
def print_status(room_a, room_b, position):
    print("\n--- Vacuum Cleaner ---")
    print("Room A:", room_a)
    print("Room B:", room_b)
    print("Vacuum Position:", position)
```

```
def vacuum_cleaner(room_a, room_b, position):
    print_status(room_a, room_b, position)
```

```
if position == 'A':
    if room_a == 'Dirty':
        print("Room A is Dirty -> SUCK")
        room_a = 'Clean'
    else:
        print("Room A is Clean -> IDLE")
```

```
print("Moving from A to B")
position = 'B'

if room_b == 'Dirty':
    print("Room B is Dirty -> SUCK")
    room_b = 'Clean'
else:
    print("Room B is Clean -> IDLE")

else:
    if room_b == 'Dirty':
        print("Room B is Dirty -> SUCK")
        room_b = 'Clean'
    else:
        print("Room B is Clean -> IDLE")

print("Moving from B to A")
position = 'A'

if room_a == 'Dirty':
    print("Room A is Dirty -> SUCK")
    room_a = 'Clean'
else:
    print("Room A is Clean -> IDLE")

print("\nFinal Status:")
print_status(room_a, room_b, position)
```

# Input

```
room_a = input("Enter status of Room A (Dirty/Clean): ")
room_b = input("Enter status of Room B (Dirty/Clean): ")
position = input("Enter vacuum position (A/B): ")
vacuum_cleaner(room_a, room_b, position)
```

```
----- RESTART: C:/Users/Student/Desktop/NITYA/vacuum cleaner.py -----
Enter status of Room A (Dirty/Clean): dirty
Enter status of Room B (Dirty/Clean): clean
Enter vacuum position (A/B): B

--- Vacuum Cleaner ---
Room A: dirty
Room B: clean
Vacuum Position: B
Room B is Clean -> IDLE
Moving from B to A
Room A is Clean -> IDLE

Final Status:

--- Vacuum Cleaner ---
Room A: dirty
Room B: clean
Vacuum Position: A
>>> |
```